

Name

Show appropriate work. Give real solutions. Do not solve linear systems or compute eigenvalues unless you need to.

1. Solve the ODE $2t y'(t) - 6y(t) = 4t^3$ with $y(1) = -3$

$$y(t) = \frac{2 \ln(t)}{t^{-3}} + \frac{C}{t^{-3}}$$

$$\text{or } y(t) = 2t^3 \ln(t) + Ct^3$$

$$\begin{aligned} y' - \frac{3}{t}y &= 2t^2 \\ \text{I.F. } v(t) &= e^{-3 \ln(t)} = t^{-3} \\ t^{-3}y' - 3t^{-4}y &= 2t^{-1} \\ (t^{-3}y)' &= 2t^{-1} \\ t^{-3}y &= \int 2t^{-1} dt + C \\ t^{-3}y &= 2 \ln(t) + C \\ \text{I.C. } 1^{-3}(-3) &= 2 \ln(1) + C \\ C &= -3 \end{aligned}$$

2. Complementary solutions y_c

- 2.1. Compute y_c for $y' = Ay$ with A below. Describe the dominant behavior as $t \rightarrow \infty$.

$$\text{In[30]: MatrixForm[Eigensystem[A = \begin{pmatrix} 6 & -15 & 15 \\ 1 & 2 & 1 \\ 1 & 11 & -8 \end{pmatrix}]]]$$

Out[30]:

$$\begin{pmatrix} -9 & 6 & 3 \\ -1, 0, 1 & 3, 1, 1 & 0, 1, 1 \end{pmatrix}$$

$$y_c(t) = C_1 y_1 + C_2 y_2 + C_3 y_3$$

$$\text{As } t \rightarrow \infty \quad y \sim C_2 e^{6t} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

exp. Growing.

$$\begin{aligned} y_1 &= e^{-9t} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \\ y_2 &= e^{6t} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \\ y_3 &= e^{3t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \end{aligned}$$

2.2. Compute y_c for $y' = A y$ with A below. Describe the dominant behavior as $t \rightarrow \infty$.

In[57]: MatrixForm[Eigensystem[A = $\begin{pmatrix} 1 & -6 \\ 3 & -5 \end{pmatrix}$]]

Out[57]: MatrixForm[
 $\begin{pmatrix} -2+3i & -2-3i \\ \{1+i, 1\} & \{1-i, 1\} \end{pmatrix}$

$$y_c(t) = C_1 y_1 + C_2 y_2$$

As $t \rightarrow \infty$ decays like e^{-2t}
 oscill. like $\cos(3t)$

$$a = -2 \quad b = 3$$

$$p = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad q = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$y_1 = e^{-2t} (\cos(3t)p - \sin(3t)q)$$

$$y_2 = e^{-2t} (\sin(3t)p + \cos(3t)q)$$

3. Compute y_c for $y' = A y$ with A below. Describe the dominant behavior as $t \rightarrow \infty$.

In[58]:

MatrixForm[Eigensystem[A = $\begin{pmatrix} 1 & -1 \\ 9 & -5 \end{pmatrix}$]]

Out[58]: MatrixForm[

$\begin{pmatrix} -2 & -2 \\ \{1, 3\} & \{0, 0\} \end{pmatrix}$

$$y_c(t) = C_1 y_1 + C_2 y_2$$

As $t \rightarrow \infty$

$$y_c \sim C_2 y_2$$

slightly slower decay than e^{-2t}

also fine to say

$$y_c \sim C_2 t e^{-2t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$y_1 = e^{-2t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$y_2 = t e^{-2t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} + e^{-2t} u$$

u satisfies

$$(A + 2I)u = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\left[\begin{array}{cc|c} 3 & -1 & 1 \\ 9 & -3 & 3 \end{array} \right] \sim \left[\begin{array}{cc|c} 3 & -1 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

second comp free.
 choose it zero!

$$u = \begin{pmatrix} 1/3 \\ 0 \end{pmatrix}$$

4. Particular solutions y_p 4.1. Write down the y_p form for $y' = Ay + b$ with $b = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$.

$$y_p(t) = u$$

Write down the Lin System for the coefficients

$$-Au = b$$

$$[-A \mid b] \text{ Aug mit Version.}$$

$$\begin{aligned} y_p &= u & 0 &= Au + b \\ y_p' &= 0 & -Au &= b \end{aligned}$$

4.2. Write down the y_p form for $y' = Ay + b$ with $b = e^{-2t} \begin{pmatrix} 3 \\ 7 \end{pmatrix}$.

$$y_p(t) = ue^{-2t}$$

Write down the Lin System for the coefficients

$$-Au - 2u = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

Aug
mit
Version

$$[-A - 2I \mid \begin{pmatrix} 3 \\ 7 \end{pmatrix}]$$

$$\begin{aligned} y_p &= ue^{-2t} \\ y_p' &= -2ue^{-2t} \\ -2ue^{-2t} &= Aue^{-2t} + e^{-2t} \begin{pmatrix} 3 \\ 7 \end{pmatrix} \\ -2u &= Au + \begin{pmatrix} 3 \\ 7 \end{pmatrix} \\ -Au - 2u &= \begin{pmatrix} 3 \\ 7 \end{pmatrix} \\ (-2 - 2I)u &= \begin{pmatrix} 3 \\ 7 \end{pmatrix} \end{aligned}$$

5. Particular solutions y_p

5.1. Write down the y_p form for $y' = Ay + b$ with $b = \begin{pmatrix} 1+t^2 \\ t \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t^2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$y_p(t) = u_0 + t u_1 + t^2 u_2$$

5.2. Write down the y_p form for $y' = Ay + b$ with $b = \cos(4t) \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

$$y_p(t) = \cos(4t) u_1 + \sin(4t) u_2$$

Write down the Lin System for the coefficients

$$\begin{bmatrix} -4I & -A & | & 0 \\ -A & 4I & | & \frac{1}{2} \end{bmatrix}$$

Any
mat
version

$$\begin{aligned} y_p &= \cos(4t) u_1 + \sin(4t) u_2 \\ y_p' &= -4 \sin(4t) u_1 + 4 \cos(4t) u_2 \\ -4 \sin(4t) u_1 + 4 \cos(4t) u_2 &= \sin(4t) A u_2 + \cos(4t) A u_1 + \cos(4t) \begin{pmatrix} 1 \\ 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{so} \quad -4 u_1 &= A u_2 \\ 4 u_2 &= A u_1 + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} -4I u_1 - A u_2 &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ -A u_1 + 4I u_2 &= \begin{pmatrix} 1 \\ 2 \end{pmatrix} \end{aligned}$$

$$\begin{bmatrix} -4I & -A \\ -A & 4I \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$